

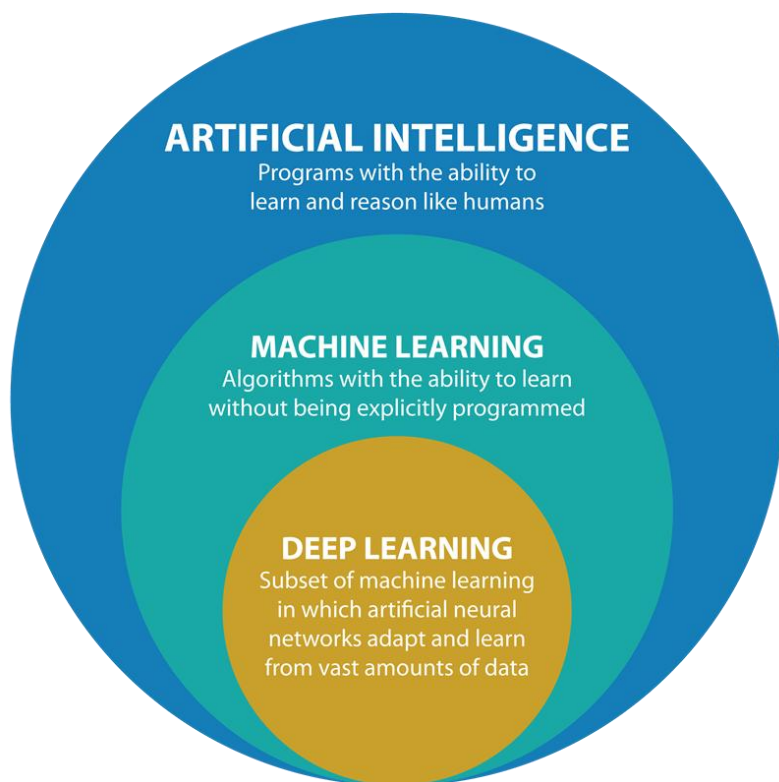
1. Python 기초 : 초보자를 위한 파이썬 300제

<https://wikidocs.net/book/922>

2. Google Colab 설치 및 UI 설명

3. Google Colab에서 python 예제 실습

Artificial Intelligence & Machine Learning



- **강인공지능/일반인공지능(AGI)**

인간과 동일한 수준 또는 그 이상의 수준으로 특정분야에 국한되지 않고 모든 분야에서 독자적으로 수행되는 인공지능

- **약인공지능**

어떤 특정한 분야의 주어진 일을 인간의 지시에 따라 제한적으로 수행하는 인공지능

Artificial Intelligence & Machine Learning

● 지도 학습

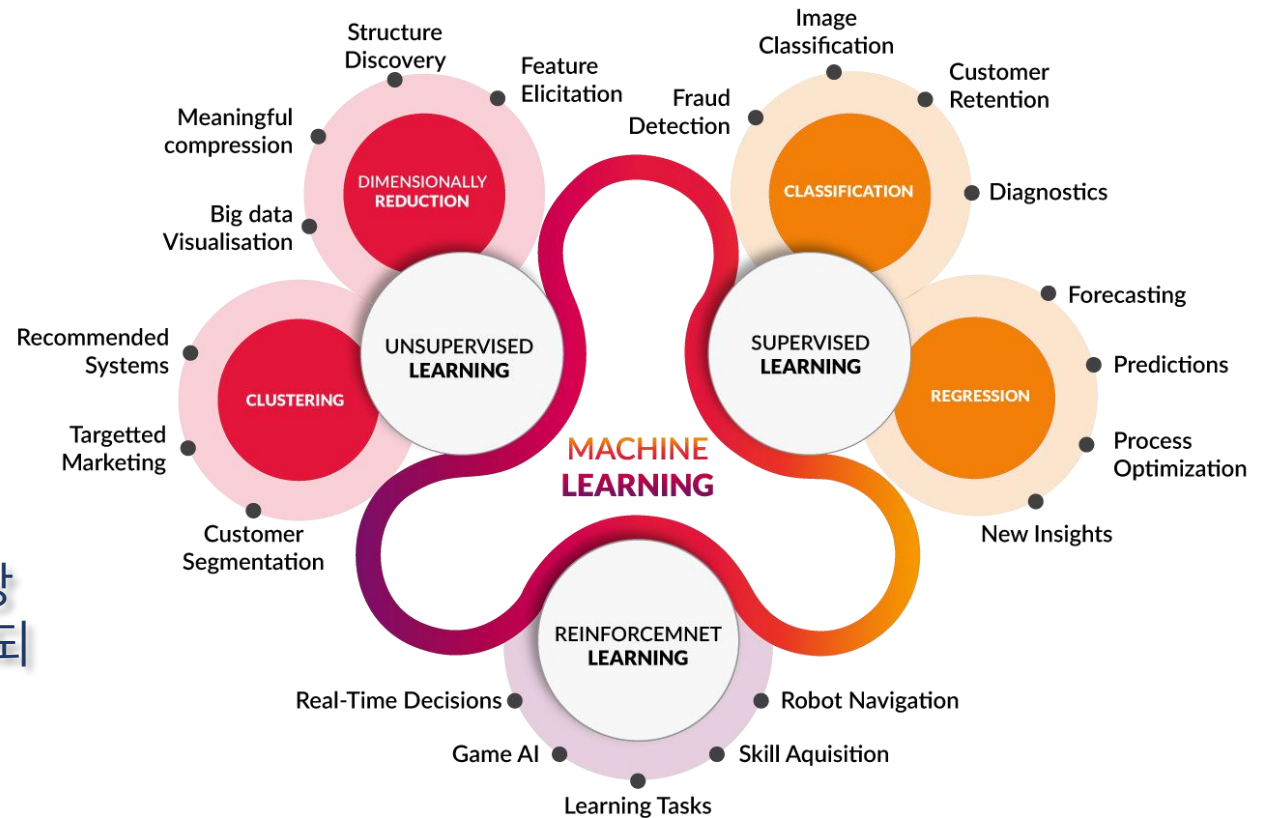
- 정답이 있는 데이터를 학습하는 것
- 분류, 회귀 등

● 비지도 학습

- 정답이 없는 데이터를 학습하는 것
- 군집화, 차원 축소 등

● 강화 학습

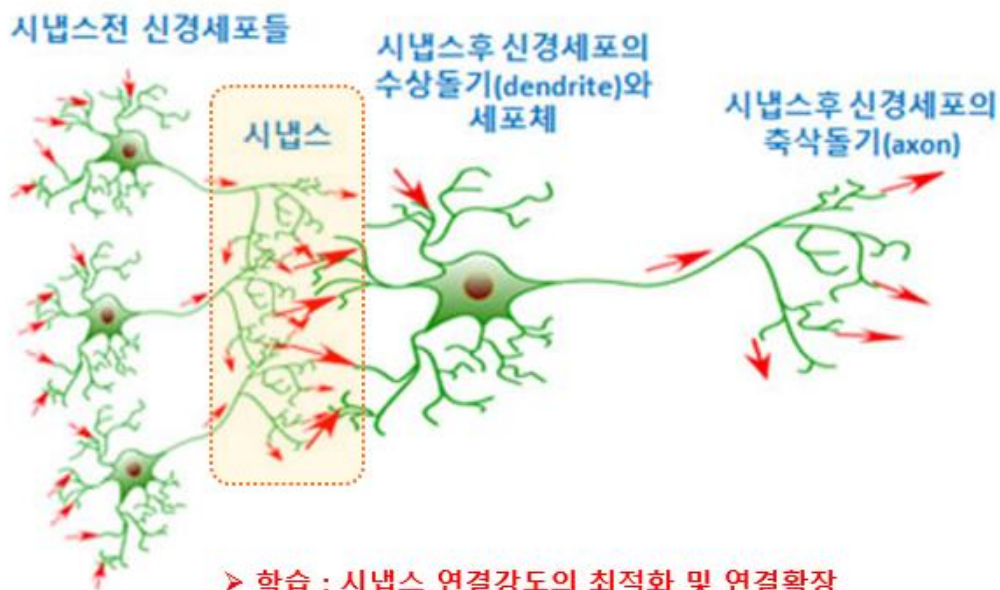
- 학습을 통해 바람직한 결과를 도출하면 보상을 얻도록 함으로써 스스로 보상이 최대가 되는 방법을 찾는 것
- 게임 등



What is machine learning? And what does AI learn?

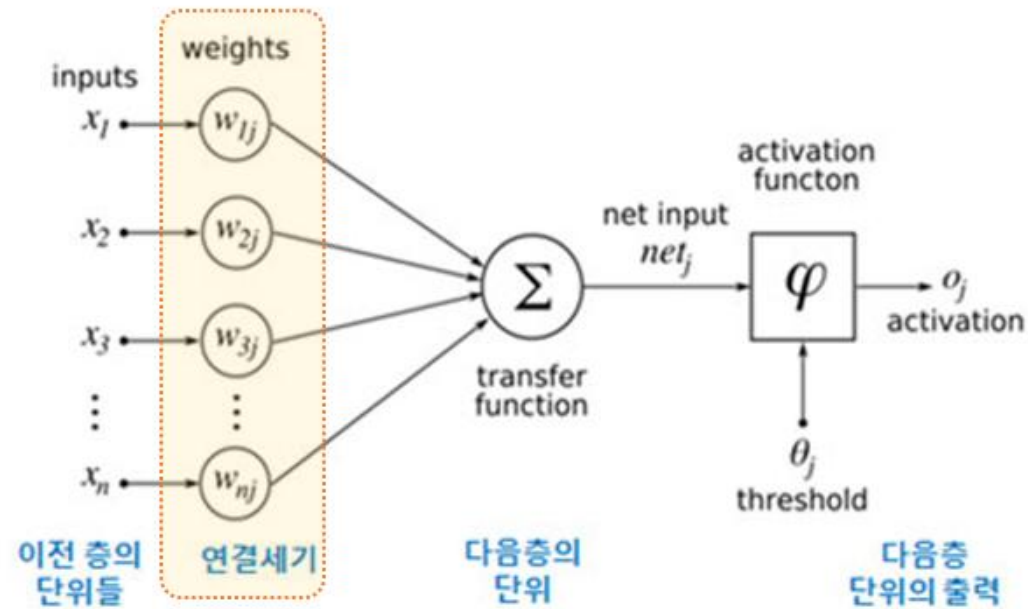


Biological Neural Network



➤ 학습 : 시냅스 연결강도의 최적화 및 연결확장

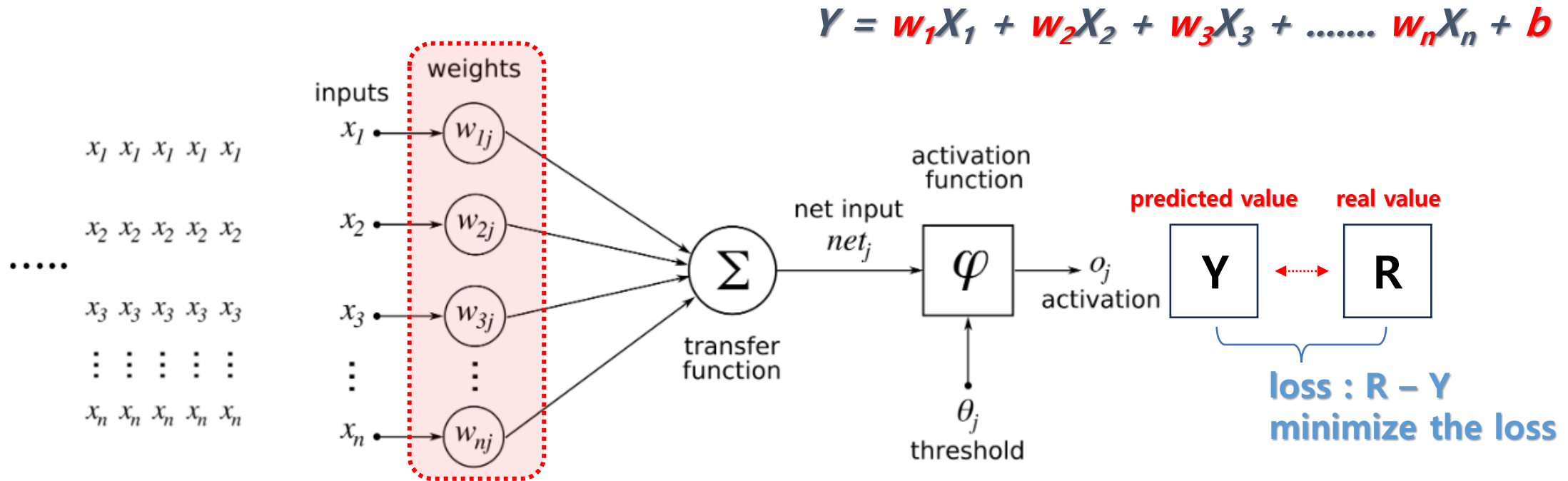
Artificial Neural Network



➤ 기계학습(ML) : weights(연결강도)를 스스로 최적화

❖ AI/ML의 기본 개념과 원리

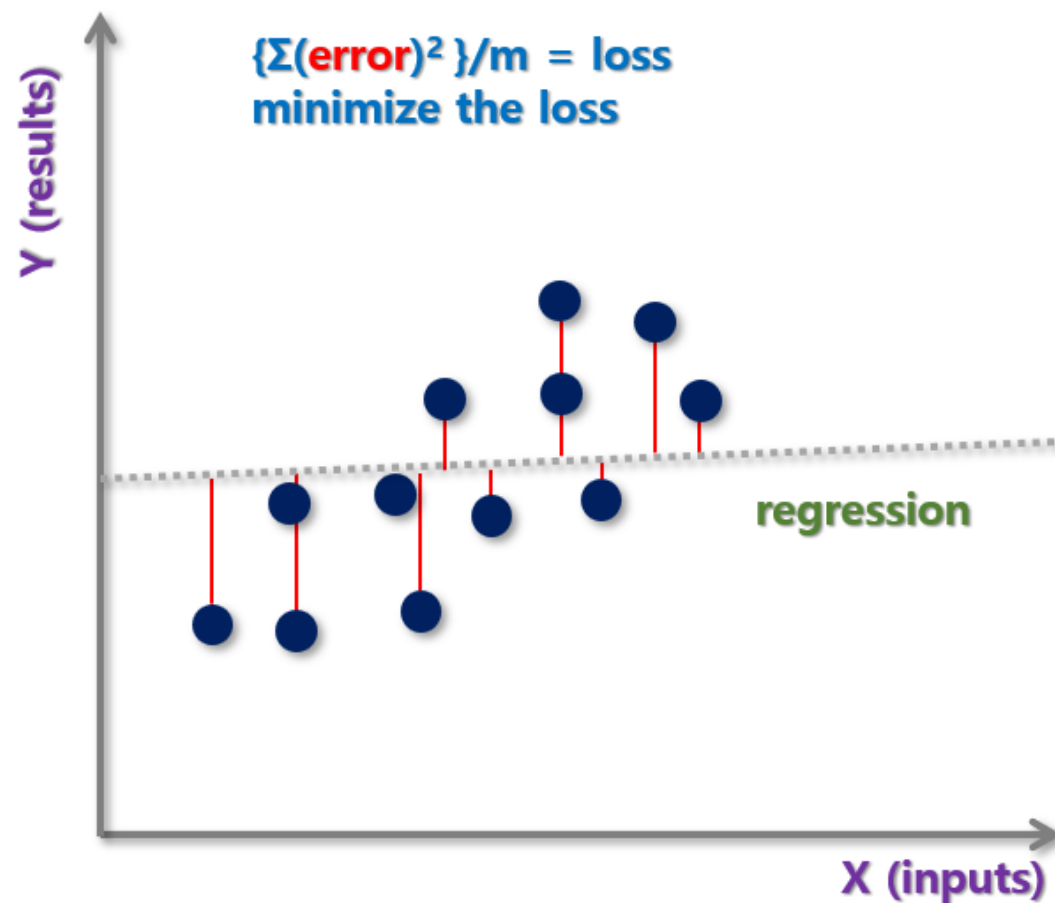
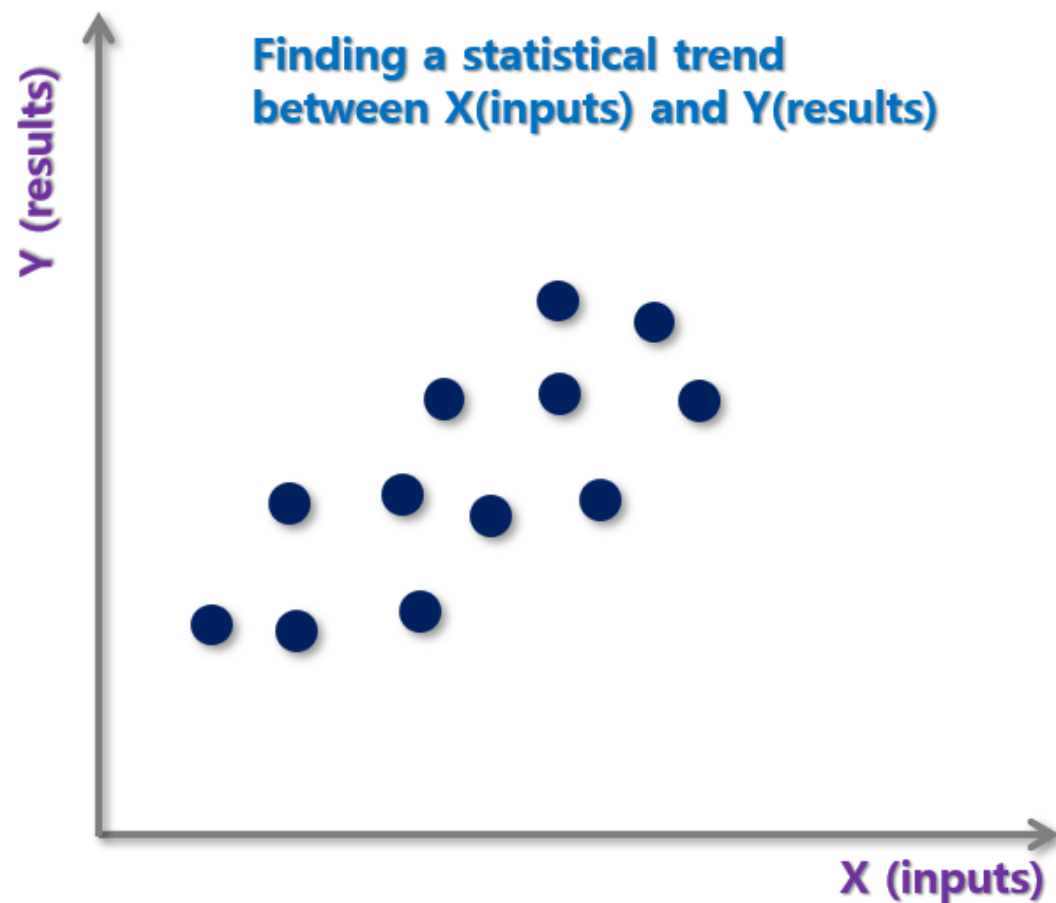
❖ Artificial Neural Network



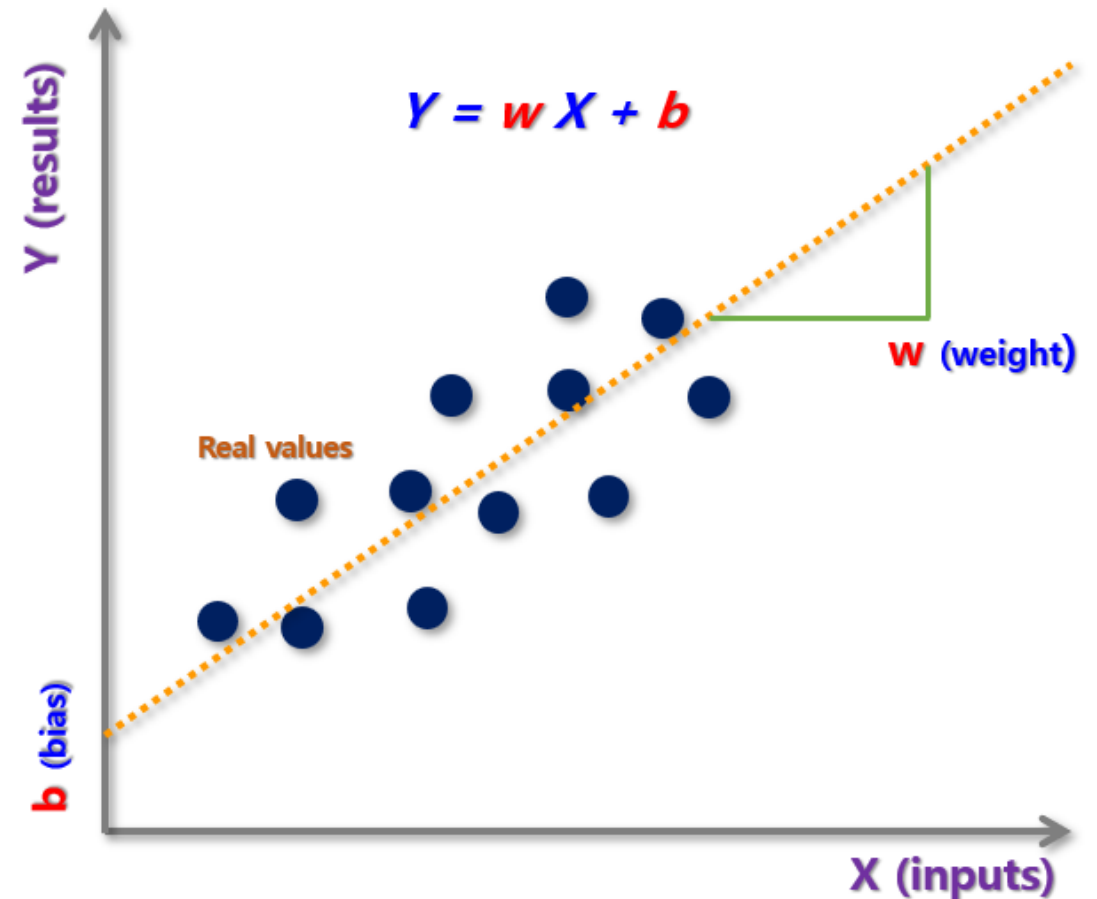
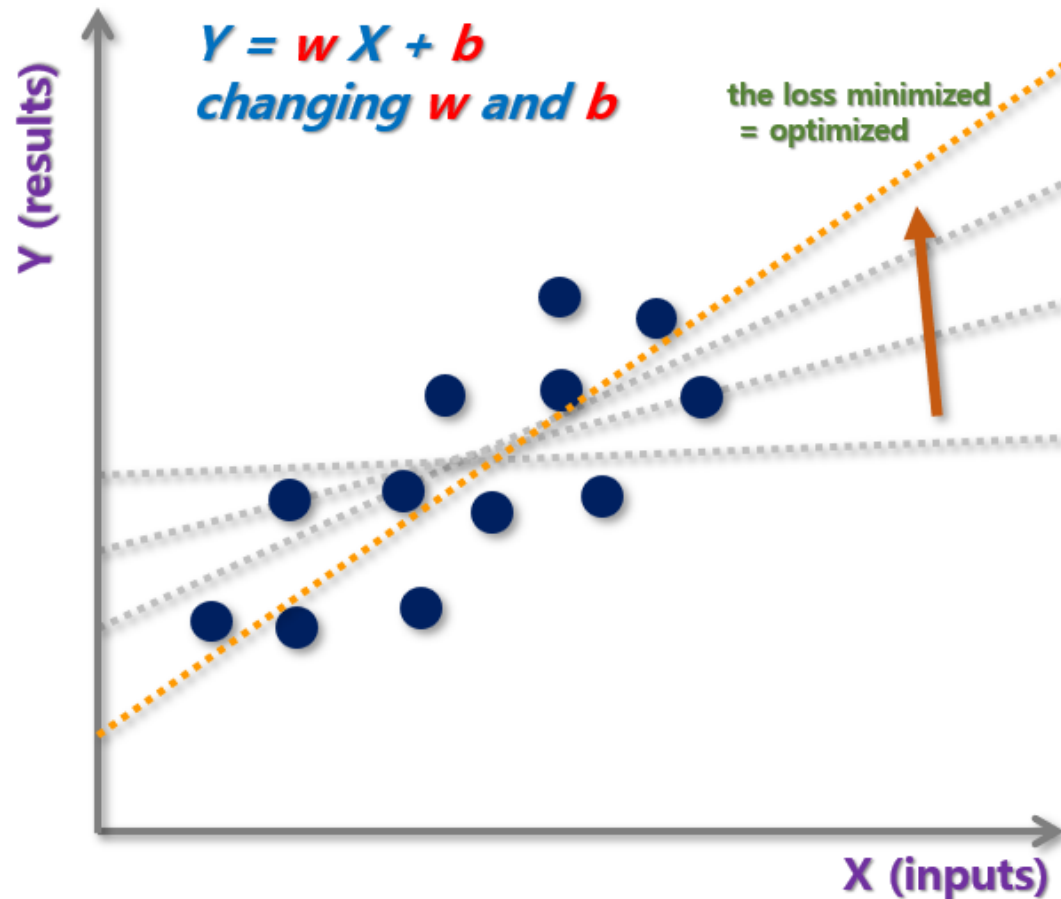
➤ 머신러닝(ML) : 연결강도(weights)를 스스로 최적화

➤ 많은 입력데이터(inputs, x)가 복잡하게 작용하여 나타난 실제 결과(R)와 그 결과를 예측한 값(Y)이 최대한 일치하도록 가중치(weights)를 스스로 최적화

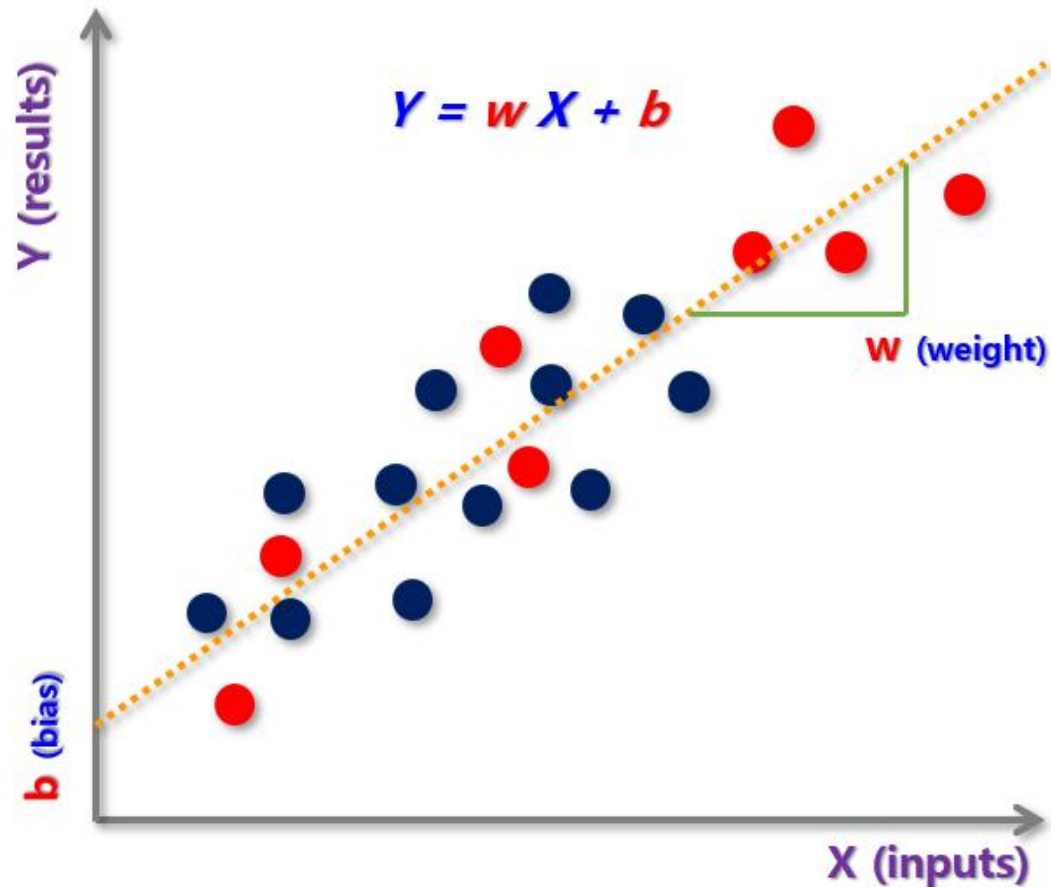
What is machine learning? And what does AI learn?



What is machine learning? And what does AI learn?



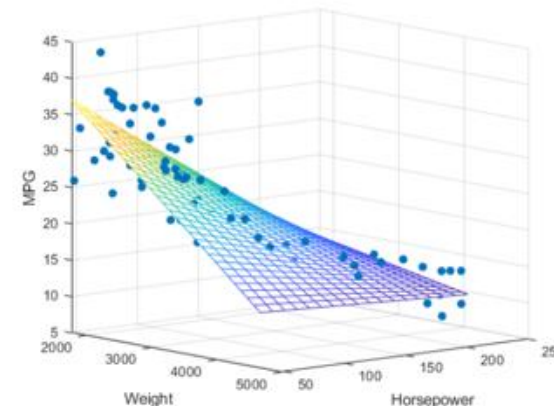
What is machine learning? And what does AI learn?



❖ Machine Learning (ML)

$$Y = w_1X_1 + w_2X_2 + w_3X_3 + \dots w_nX_n + b$$

- ✓ to find optimized weight-set ($w_1, \dots w_n$) and bias (b) from relationship between **multiple** X and single Y
- ✓ possible to predict reasonable RESULTS from untrained test-set using pre-optimized weights and bias



What does AI learn from? ... *Big Data*

Group	Dye/Color	Allyl	Marking	C	H	N	O	S	Cl	Br	NW	Radius(atomic)	SL Vol(A3)	Hard Vol	Quality	Rot Bond	Aromatic Area(A2)	Polar Area(A2)	H bond Acceptor	H bond Donors	Phenyl	Alkyl/Chain	Total allyl	logP	Molar E	MP	QV	Stokes shift	real_K5
Asn	Blue	butyl	train	20.00	27.00	3.00	0.00	0.00	0.00	0.00	363.23	7.000	293.758	221.993	1.000	293.758	0.000	2.148	4.000	7.000	7.000	7.000	12.000	6.000	12.000	6.000	12.000	6.000	27.1
Asn	Br/Yellow	heptyl	train	24.00	35.00	3.00	0.00	0.00	0.00	0.00	363.263	8.000	363.263	224.293	1.000	13.000	754.903	27.960	3.000	0.000	2.000	4.000	12.000	6.000	12.000	6.000	12.000	6.000	28.3
Disazo	Di/Yellow	methoxy	train	19.00	16.00	4.00	1.00	0.00	0.00	0.00	316.36	8.000	244.265	244.265	1.594	8.000	566.872	69.670	5.000	1.000	3.000	1.000	1.000	7.004	30000	145	0.00	10.0	
Disazo	Di/Yellow	ethyl	train	20.00	18.00	4.00	1.00	0.00	0.00	0.00	330.38	8.000	261.650	261.650	1.611	5.000	595.135	69.670	5.000	1.000	3.000	2.000	2.000	7.422	30000	146	0.00	10.5	
Disazo	Di/Yellow	propyl	train	21.00	20.00	4.00	1.00	0.00	0.00	0.00	344.41	9.000	278.997	278.997	1.631	6.000	636.229	69.670	5.000	1.000	3.000	3.000	3.000	7.636	30000	145	0.00	14.7	
Disazo	Di/Yellow	butyl	train	22.00	22.00	4.00	1.00	0.00	0.00	0.00	358.44	9.000	296.358	261.650	1.658	7.000	657.417	69.670	5.000	1.000	3.000	4.000	4.000	8.256	30000	130	0.00	14.9	
Disazo	Di/Yellow	pentyl	train	23.00	24.00	4.00	1.00	0.00	0.00	0.00	372.46	10.000	313.718	261.650	1.681	8.000	686.618	69.670	5.000	1.000	3.000	5.000	5.000	8.674	30000	125	0.00	15.2	
Disazo	Di/Yellow	heptyl	train	24.00	26.00	4.00	1.00	0.00	0.00	0.00	386.49	10.000	331.091	261.650	1.701	9.000	719.640	69.670	5.000	1.000	3.000	6.000	6.000	9.091	30000	120	0.00	15.5	
Disazo	Di/Yellow	heptyl	train	25.00	28.00	4.00	1.00	0.00	0.00	0.00	400.52	11.000	348.471	261.650	1.725	10.000	751.082	69.670	5.000	1.000	3.000	7.000	7.000	9.508	30000	115	0.00	15.0	
Asn	Red	methoxy	train	14.00	13.00	4.00	2.00	0.00	1.00	0.00	304.07	7.000	223.100	223.100	1.507	4.000	504.049	79.770	4.000	0.000	2.000	1.000	2.000	4.655	30000	120	0.00	8.3	
Asn	Red	ethyl	train	16.00	17.00	4.00	2.00	0.00	1.00	0.00	332.78	7.000	258.122	258.122	1.532	6.000	551.492	79.770	4.000	0.000	2.000	2.000	4.000	5.037	30000	100	0.00	16.7	
Asn	Red	propyl	train	18.00	21.00	4.00	2.00	0.00	1.00	0.00	360.14	8.000	292.622	258.122	1.564	8.000	615.603	79.770	4.000	0.000	2.000	3.000	4.000	5.949	30000	90	0.00	24.8	
Asn	Red	butyl	train	20.00	25.00	4.00	2.00	0.00	1.00	0.00	386.17	8.000	327.565	258.122	1.632	10.000	677.718	79.770	4.000	0.000	2.000	4.000	6.000	6.861	30000	85	0.00	27.2	
Asn	Red	heptyl	train	24.00	33.00	4.00	2.00	0.00	1.00	0.00	444.23	9.000	397.146	258.122	1.725	14.000	862.548	79.770	4.000	0.000	2.000	6.000	12.000	8.665	30000	80	0.00	28.3	
AQ	Blue	methoxy	train	16.00	14.00	2.00	2.00	0.00	0.00	0.00	266.29	5.000	201.254	201.254	1.431	2.000	447.876	58.200	4.000	2.000	3.000	1.000	2.000	1.413	18000	220	0.00	1.2	
AQ	Blue	ethyl	train	18.00	18.00	2.00	2.00	0.00	0.00	0.00	284.14	5.000	235.546	235.546	1.496	4.000	513.185	58.200	4.000	2.000	3.000	2.000	4.000	2.689	18000	215	0.00	2.0	
AQ	Blue	butyl	train	22.00	26.00	2.00	2.00	0.00	0.00	0.00	350.45	7.000	305.060	235.546	1.605	8.000	636.226	58.200	4.000	2.000	3.000	4.000	6.000	3.896	18000	120	0.00	12.1	
AQ	Blue	pentyl	train	24.00	30.00	2.00	2.00	0.00	0.00	0.00	376.51	8.000	339.603	235.546	1.654	10.000	700.690	58.200	4.000	2.000	3.000	5.000	10.000	4.730	18000	112	0.00	15.1	
AQ	Blue	heptyl	train	26.00	34.00	2.00	2.00	0.00	0.00	0.00	406.56	9.000	374.545	235.546	1.701	12.000	763.133	58.200	4.000	2.000	3.000	6.000	12.000	5.565	18000	105	0.00	18.5	
AQ	Blue	heptyl	train	26.00	38.00	2.00	2.00	0.00	0.00	0.00	434.61	10.000	409.325	235.546	1.745	14.000	825.637	58.200	4.000	2.000	3.000	7.000	14.000	6.399	18000	100	0.00	17.0	
AQ	Blue	octyl	train	30.00	42.00	2.00	2.00	0.00	0.00	0.00	462.67	11.000	444.089	235.546	1.787	16.000	888.117	58.200	4.000	2.000	3.000	8.000	16.000	7.234	18000	95	0.00	16.4	
AQ	Blue	butyl	train	18.00	17.00	1.00	3.00	0.00	0.00	0.00	295.33	6.000	233.902	199.162	1.490	4.000	528.523	66.400	4.000	2.000	3.000	4.000	6.000	2.768	18000	112	0.00	6.3	
AQ	Blue	heptyl	train	24.00	27.00	1.00	3.00	0.00	0.00	0.00	446.54	7.000	361.279	199.162	1.629	4.000	717.684	66.400	4.000	2.000	3.000	6.000	10.000	4.855	18000	100	0.00	13.4	
AQ	Blue	heptyl	train	21.00	23.00	1.00	3.00	0.00	0.00	0.00	337.41	7.000	286.033	199.162	1.574	7.000	602.219	66.400	4.000	2.000	3.000	7.000	8.000	4.030	18000	88	0.00	13.2	
AQ	Blue	octyl	train	22.00	25.00	1.00	3.00	0.00	0.00	0.00	351.44	8.000	303.421	199.162	1.599	8.000	633.459	66.400	4.000	2.000	3.000	8.000	10.000	4.437	18000	85	0.00	13.2	
AQ	Blue	decyl	train	24.00	29.00	1.00	3.00	0.00	0.00	0.00	379.49	9.000	338.179	199.162	1.649	10.000	695.929	66.400	4.000	2.000	3.000	10.000	10.000	5.272	18000	72	0.00	14.2	
AQ	Blue	dodecyl	train	26.00	33.00	1.00	3.00	0.00	0.00	0.00	407.55	10.000	372.941	199.162	1.696	12.000	758.395	66.400	4.000	2.000	3.000	12.000	12.000	6.106	18000	71	0.00	14.4	
Asn	Orange	ethyl	train	18.00	16.00	2.00	1.00	0.00	0.00	0.00	276.33	6.000	222.350	222.350	1.498	3.000	497.292	44.950	3.000	1.000	3.000	2.000	2.000	5.903	30000	120	0.00	1.8	
Asn	Orange	butyl	train	20.00	20.00	2.00	1.00	0.00	0.00	0.00	304.39	7.000	257.063	222.350	1.553	5.000	559.611	44.950	3.000	1.000	3.000	4.000	4.000	6.817	30000	100	0.00	2.0	
Asn	Orange	heptyl	train	22.00	24.00	2.00	1.00	0.00	0.00	0.00	332.44	8.000	291.829	222.350	1.605	7.000	622.104	44.950	3.000	1.000	3.000	6.000	6.000	7.652	30000	90	0.00	22.2	
Asn	Orange	octyl	train	24.00	28.00	2.00	1.00	0.00	0.00	0.00	360.49	9.000	326.572	222.350	1.654	9.000	684.547	44.950	3.000	1.000	3.000	8.000	8.000	8.496	30000	85	0.00	21.6	
Asn	Orange	decyl	train	26.00	32.00	2.00	1.00	0.00	0.00	0.00	388.55	10.000	361.310	222.350	1.708	11.000	746.999	44.950	3.000	1.000	3.000	10.000	10.000	9.321	30000	80	0.00	21.6	
AQ	Alcohol/Magenta	butyl	train	18.00	17.00	1.00	1.00	0.00	0.00	0.00	240.34	6.000	140.534	140.534	1.503	4.000	320.444	69.620	5.000	2.000	3.000	4.000	4.000	2.142	18000	172	0.00	5.9	
AQ	Alcohol/Magenta	heptyl	train	20.00	21.00	1.00	1.00	0.00	0.00	0.00	339.39	7.000	275.268	205.854	1.558	6.000	582.510	69.620	5.000	2.000	3.000	6.000	6.000	3.177	18000	130	0.00	6.0	
AQ	Alcohol/Magenta	heptyl	train	21.00	23.00	1.00	1.00	0.00	0.00	0.00	353.41	8.000	292.665	205.854	1.565	7.000	613.732	69.620	5.000	2.000	3.000	7.000	7.000	3.594	18000	116	0.00	6.0	
AQ	Alcohol/Magenta	heptyl	train	22.00	25.00	1.00	1.00	0.00	0.00	0.00	367.46	9.000	310.036	205.854	1.573	8.000	645.876	69.620	5.000	2.000	3.000	8.000	8.000	4.011	18000	110	0.00	6.1	
AQ	Alcohol/Magenta	heptyl	train	23.00	27.00	1.00	1.00	0.00	0.00	0.00	381.46	9.000	327.432	205.854	1.635	9.000	676.189	69.620	5.000	2.000	3.000	9.000	9.000	4.429	18000	108	0.00	8.9	
AQ	Alcohol/Magenta	decyl	train	24.00	29.00	1.00	1.00	0.00	0.00	0.00	395.49	9.000	344.778	205.854	1.659	10.000	707.405	69.620	5.000	2.000	3.000	10.000	10.000	4.846	18000	98	0.00	8.2	
AQ	Alcohol/Magenta	dodecyl	train	26.00	33.00	1.00	1.00	0.00	0.00	0.00	423.54	10.000	379.542	205.854	1.699	12.000	799.915	69.620	5.000	2.000	3.000	12.000	12.000	5.681	18000	84	0.00	8.2	
AQ	Green	butyl	train	30.00	26.00	2.00	2.00	0.00	0.00	0.00	446.54	6.000	361.279	199.162	1.629	4.000	717.684	58.200	4.000	2.000	3.000	6.000	7.158	18000	290	0.00	1.2		
AQ	Green	propyl	train	32.00	30.00	2.00	2.00	0.00	0.00	0.00	474.59	10.000	396.060	199.162	1.705	8.000	780.083	58.200	4.000	2.000	3.000	8.000	7.993	18000	150	0.00	5.2		
AQ	Green	butyl	train	34.00	34.00	2.00	2.00	0.00	0.00	0.00	502.65	11.000	430.795	199.162	1.749	10.000	842.459	58.200	4.000	2.000	3.000	10.000	8.600	8.827	15000	145	0.00	6.1	
AQ	Green	pentyl	train	36.00	36.00	2.00	2.00	0.00	0.00	0.00	530.70	12.000	465.564	199.162	1.791	12.000	904.871	58.200	4.000	2.000	3.000	12.000	10.000	9.662	15000	124	0.00	5.4	
AQ	Green	heptyl	train	38.00	42.																								

What is machine learning? And what does AI learn?

Big data : m data with n X

$$\begin{aligned}Y_1 &= w_1X_{11} + w_2X_{12} + w_3X_{13} + \dots w_nX_{1n} + b_1 \\Y_2 &= w_1X_{21} + w_2X_{22} + w_3X_{23} + \dots w_nX_{2n} + b_1 \\Y_3 &= w_1X_{31} + w_2X_{32} + w_3X_{33} + \dots w_nX_{3n} + b_1 \\&\vdots \\Y_m &= w_1X_{m1} + w_2X_{m2} + w_3X_{m3} + \dots w_nX_{mn} + b_1\end{aligned}$$

Real results - Predicted results = error

$$\begin{aligned}R_1 - Y_1 &= error_1 \rightarrow error_1^2 \\R_2 - Y_2 &= error_2 \rightarrow error_2^2 \\R_3 - Y_3 &= error_3 \rightarrow error_3^2 \\&\vdots \\R_m - Y_m &= error_m \rightarrow error_m^2\end{aligned}$$

$$\Sigma(error)^2 \rightarrow$$

$$\{\Sigma(error)^2\}/m : LOSS$$

1-1. Calculate from Y_1 to Y_m for $X_{11} \sim X_{mn}$
using a randomly determined initial set of weights and bias

1-2. Calculate ERROR between Y and Real value(R), and then calculate LOSS = **1st epoch**

2-1. Update weights and bias

2-2. Repeat calculations from Y_1 to Y_m for $X_{11} \sim X_{mn}$ using the updated set of weights and bias

2-3. Calculate new ERROR between Y and Real value, and then calculate new LOSS = **2nd epoch**

3-1. Update weights and bias again

3-2. Repeat this process **until the LOSS is minimized**

4-1. The pre-trained AI can predict new RESULTS(Y) for new input-data(Xs) using optimized weight-set and bias

What does AI learn from? ... *Big Data*

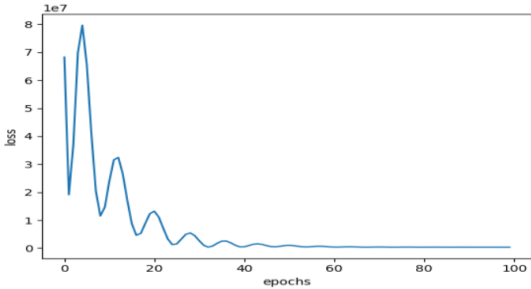
구분	M.W.	Aromaticity	Polarity	Q.Y.	λ_{em}	Planarity	OLED Lifespan
훈련용	1,100	2.2	1.3	0.10	445	0.49	10,500
	890	2.9	2.1	0.19	495	0.61	13,100
	810	4.3	4.2	0.40	565	0.65	13,500
	680	5.6	6.3	0.58	610	0.79	16,600
	580	6.2	7.9	0.82	640	0.95	17,100
	505	7.4	9.9	0.90	710	1.00	20,100
예측용	850	6.1	2.3	0.84	460	0.85	?????????

initial epoch : *randomly determined weights and bias*

W1	M.W.	W2	Aromatic	W3	Polarity	W4	Q.Y.	W5	λem	W6	Planarity	Bias	OLED Lifespan	Estimated	Error
5.87 x	1,100	5.53 x	2.2	4.53 x	1.3	4.38 x	0.10	5.66 x	445	5.11 x	0.49	+ 4.99	10,500	9,001	1,498
	890		2.9		2.1		0.19		495		0.61		13,100	8,060	5,040
	810		4.3		4.2		0.40		565		0.65		13,500	8,005	5,495
	680		5.6		6.3		0.58		610		0.79		16,600	7,515	9,085
	580		6.2		7.9		0.82		640		0.95		17,100	7,110	9,989
	505		7.4		9.9		0.90		710		1.00		20,100	7,082	13,017

$$Y_1 = w_1X_{11} + w_2X_{12} + w_3X_{13} + \dots w_nX_{1n} + b_1$$
$$Y_2 = w_1X_{21} + w_2X_{22} + w_3X_{23} + \dots w_nX_{2n} + b_1$$
$$Y_3 = w_1X_{31} + w_2X_{32} + w_3X_{33} + \dots w_nX_{3n} + b_1$$
$$\vdots$$
$$Y_m = w_1X_{m1} + w_2X_{m2} + w_3X_{m3} + \dots w_nX_{mn} + b_1$$

$$R_1 - Y_1 = error_1 \rightarrow error_1^2$$
$$R_2 - Y_2 = error_2 \rightarrow error_2^2$$
$$R_3 - Y_3 = error_3 \rightarrow error_3^2$$
$$\vdots$$
$$R_m - Y_m = error_m \rightarrow error_m^2$$



$\{\sum(error)^2\}/6 = \text{Loss} : 68,267,028$

final epoch (100) : *optimized weights and bias*

W1	M.W.	W2	Aromatic	W3	Polarity	W4	Q.Y.	W5	λem	W6	Planarity	Bias	OLED Lifespan	Estimated	Error
-1.78 x	1,100	43.40 x	2.2	55.62 x	1.3	54.25 x	0.10	27.69 x	445	35.17 x	0.49	+ 16.37	10,500	10,571	-71
	890		2.9		2.1		0.19		495		0.61		13,100	12,413	687
	810		4.3		4.2		0.40		565		0.65		13,500	14,684	-1,184
	680		5.6		6.3		0.58		610		0.79		16,600	16,350	250
	580		6.2		7.9		0.82		640		0.95		17,100	17,492	-392
	505		7.4		9.9		0.90		710		1.00		20,100	19,733	367

$\{\sum(error)^2\}/6 = \text{Loss} : 371,674$

Predict : *using the optimized weights and bias*

W1	M.W.	W2	Aromatic	W3	Polarity	W4	Q.Y.	W5	λem	W6	Planarity	Bias	OLED Lifespan
-1.78 x	850	43.40 x	6.1	55.62 x	2.3	54.25 x	0.84	27.69 x	460	35.17 x	0.85	+ 16.37	11,709

Machine Learning Example

구분	공부시간	집중시간	운동시각	알바시간	성적
훈련용	5	1	10	8	15
	2	1	8	9	21
	3	2	1	10	29
	5	4	9	6	38
	4	4	5	7	52
	5	4	6	4	63
	6	5	2	5	70
	9	7	4	1	78
	7	7	3	2	91
	9	8	3	3	95
예측용	7	6	2	9	???

Machine Learning Basics

Python ...

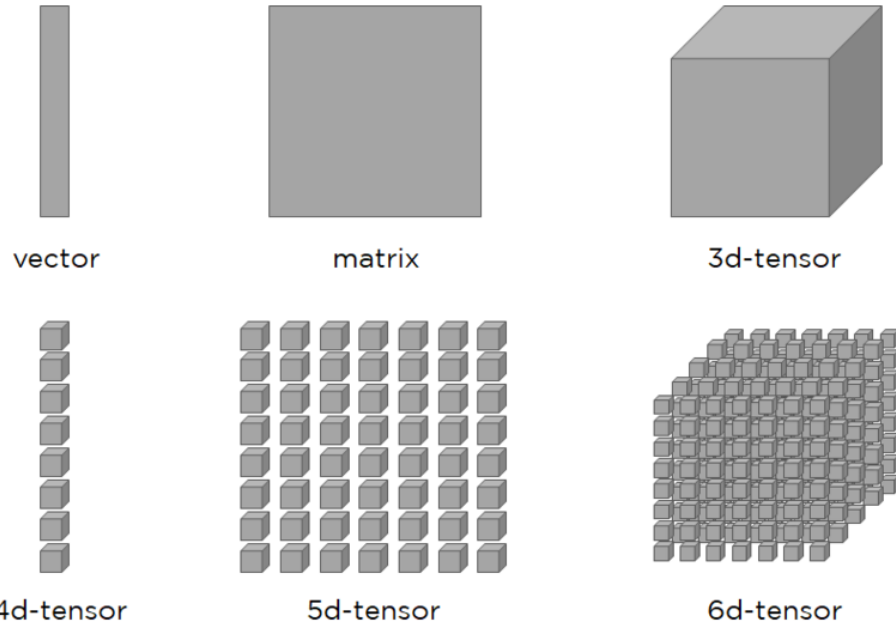
Tensorflow

Keras

Pytorch ...

Numpy

Pandas ...



RANK	TYPE	EXAMPLE
0	scalar	[1]
1	vector	[1,1]
2	matrix	[[1,1],[1,1]]
3	3-tensor	[[[1,1],[1,1]],[[1,1],[1,1]],[[1,2],[2,1]]]
n	n-tensor	

Machine Learning Basics

Model Layer

Activation function : linear, sigmoid, relu, ...

Loss/Cost function : mse, cross-entropy, ...

Optimizer : sgd, adam, ...

Learning rate

Epoch ...

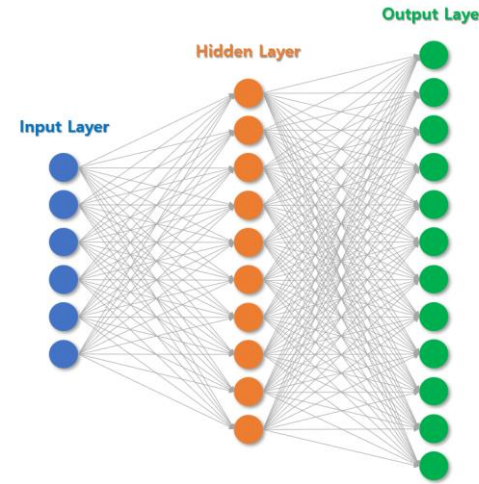
Back propagation

Parameter : weight, bias

Hyperparameter : learning rate, epoch, ...

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

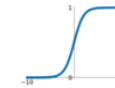
$$CELoss = - \sum_{c=1}^N y_c \log P_c$$



Activation Functions

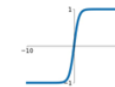
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



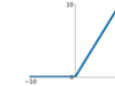
tanh

$$\tanh(x)$$



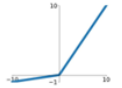
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

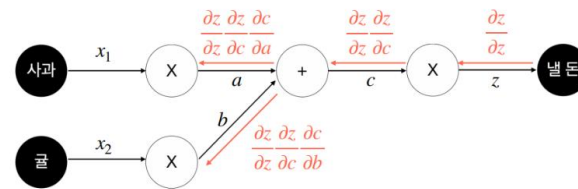
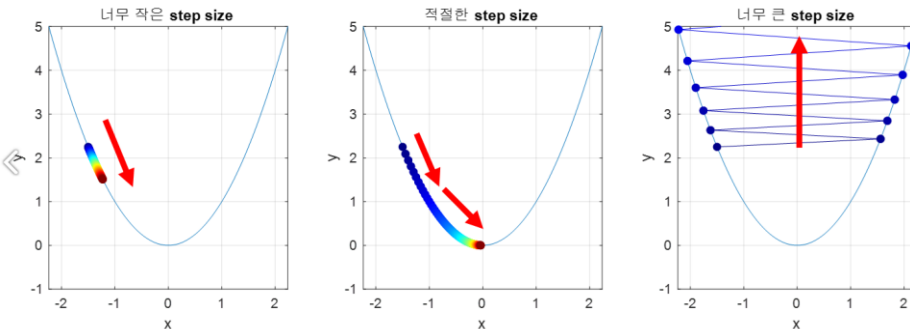
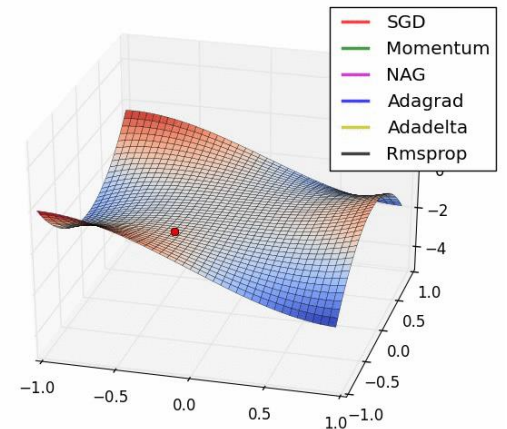
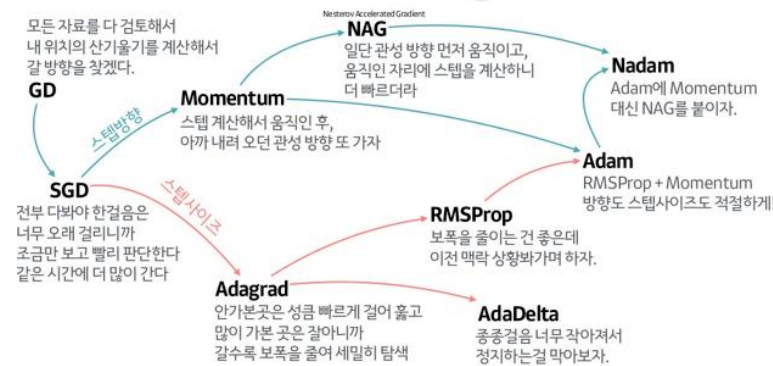
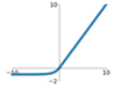


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



$$W_{t+1} = W_t - \alpha \frac{\partial L}{\partial W_t}$$

Current Parameter value

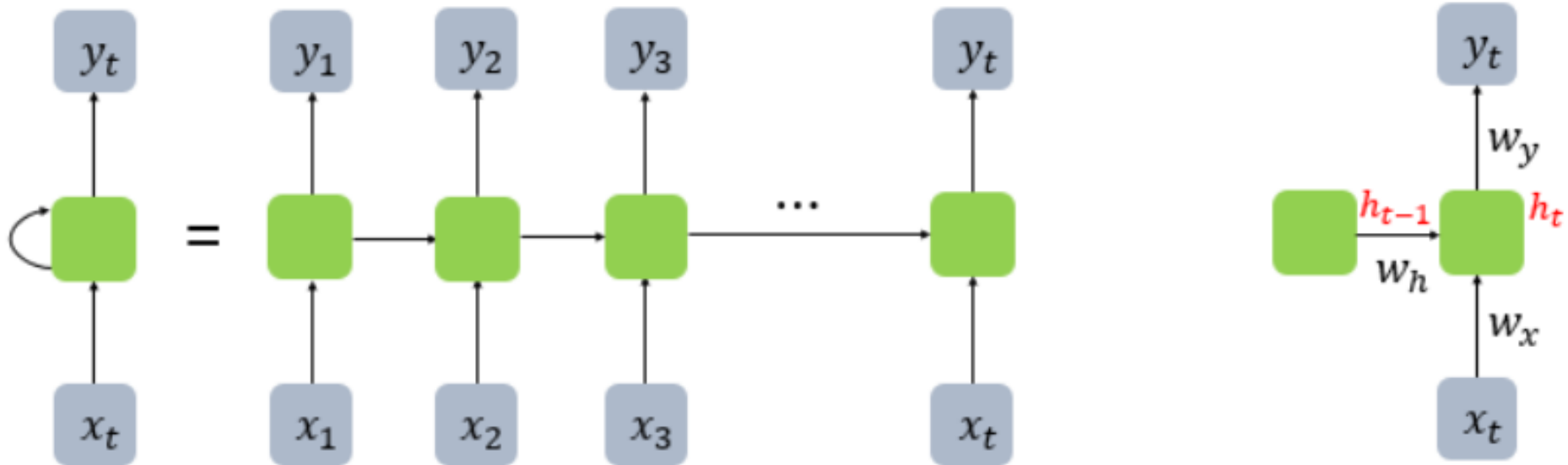
Gradient

Updated Parameter value

Learning Rate

RNN (Recurrent Neural Network)

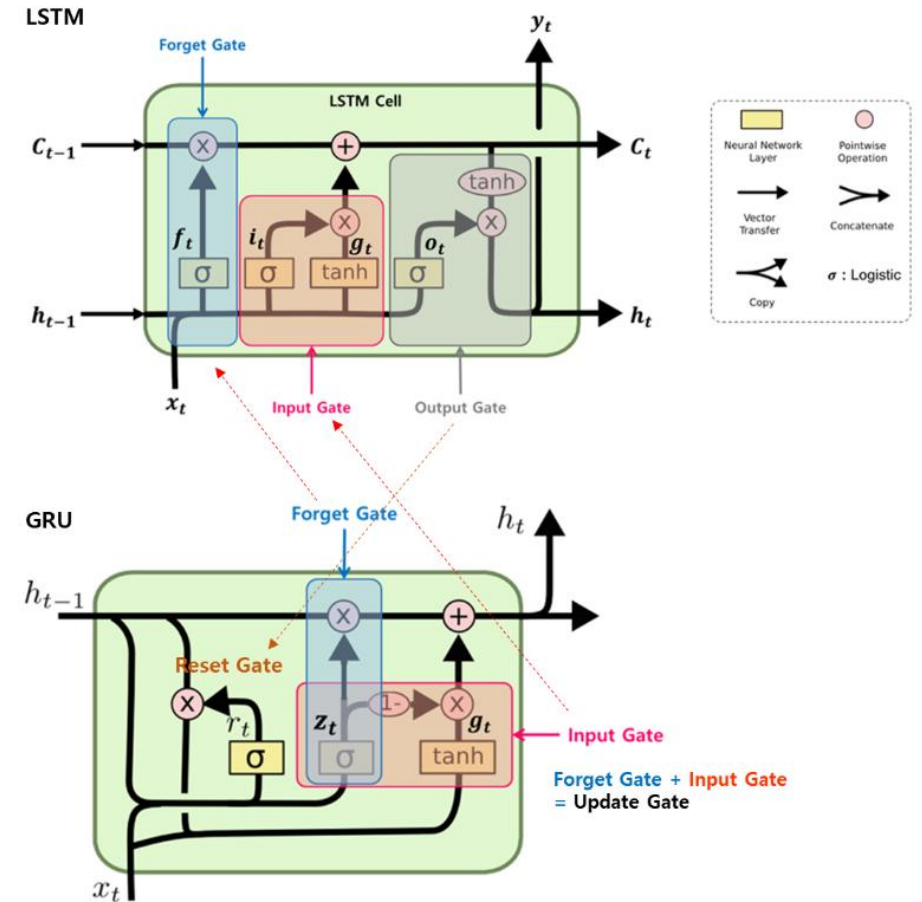
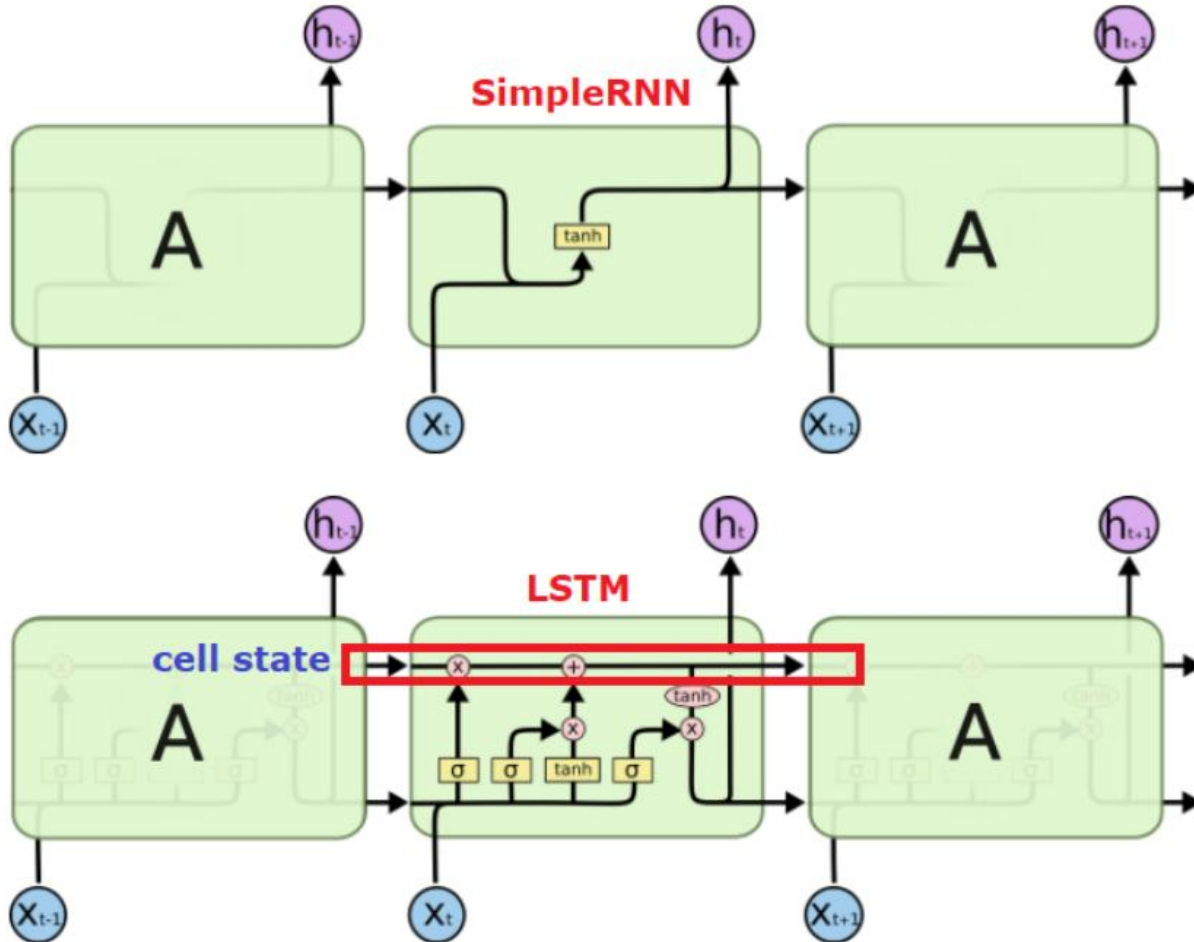
Window-size (time-steps), Horizon-factor, ...



- 은닉층 : $h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$
- 출력층 : $y_t = f(W_y h_t + b)$

- Long-term dependency
- Long-distance dependency
- Long-range dependency

LSTM (Long Short-Term Memory) & GRU (Gated Recurrent Unit)



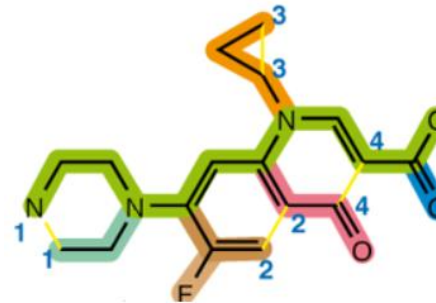
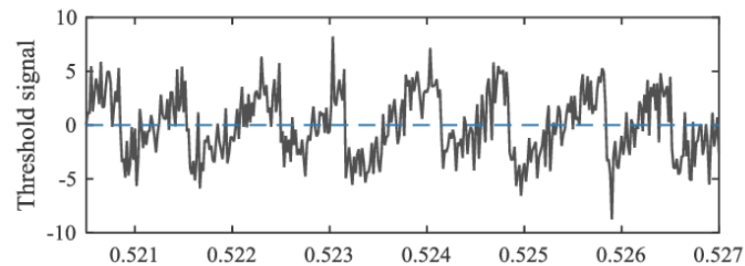
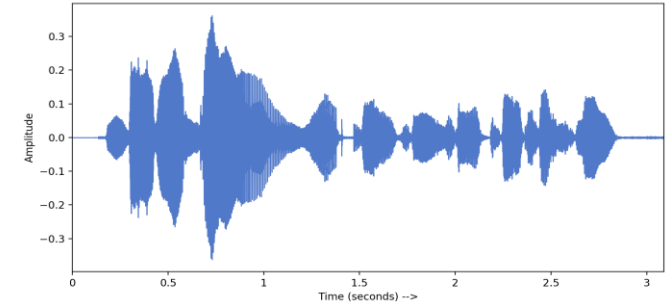
Application of RNN (Recurrent Neural Network)

Sound sequence : Voice, Music, ...

Text Sentence (→ChatGPT)

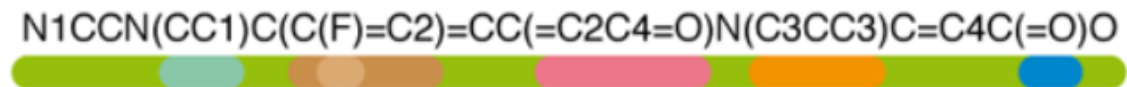
Sensor sequence signal

Molecular structure : protein folding, drug design, ...
(SMILE format, Fingerprint)



Lack of access to reading and writing put blind people at a serious

disadvantage in nineteenth-century society. Text was one of the primary methods through which people engaged with culture, communicated with others, and accessed information; without a well-developed reading system that did not rely on sight, blind people were excluded from social participation (Weygand, 2009). While



Computer Vision(CV)

Dataset : mnist, fashion mnist, cifar10, ...

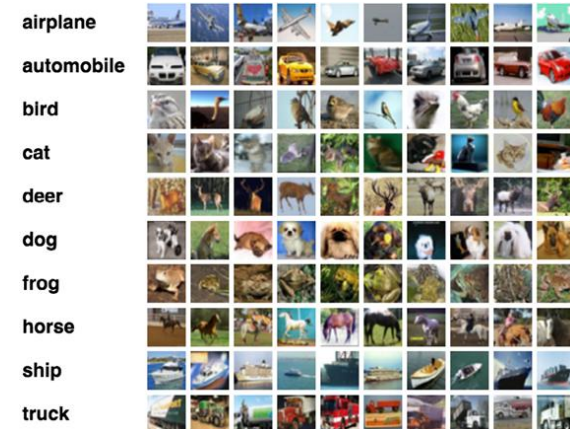
Flatten, Dense/Dropout, One-hot encoding, ...

Activation function : relu, softmax, ...

Loss function : categorical_crossentropy, ...

CV2

NN vs. CNN



Computer Vision(CV)

Dataset : mnist, fashion mnist, cifar10, ...

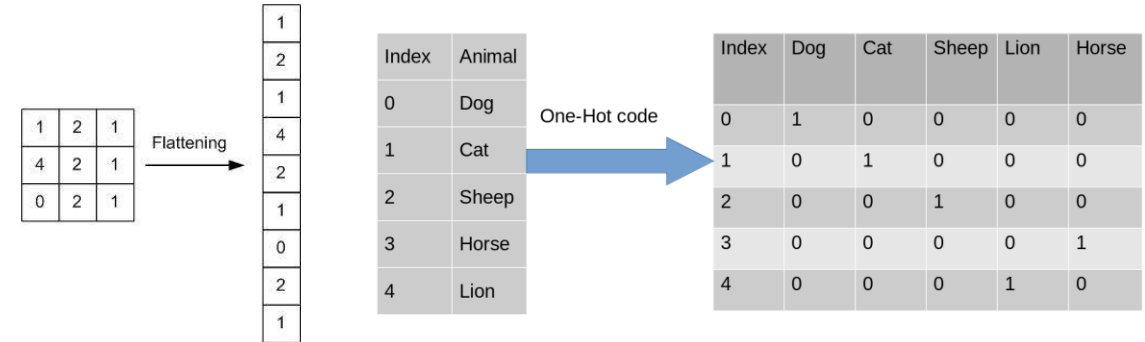
Flatten, Dense/Dropout, One-hot encoding, ...

Activation function : **relu**, **softmax**, ...

Loss function : categorical_crossentropy, ...

CV2

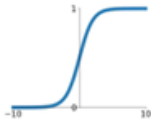
NN vs. CNN



Activation Functions

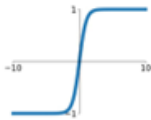
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



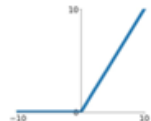
tanh

$$\tanh(x)$$



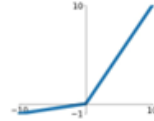
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

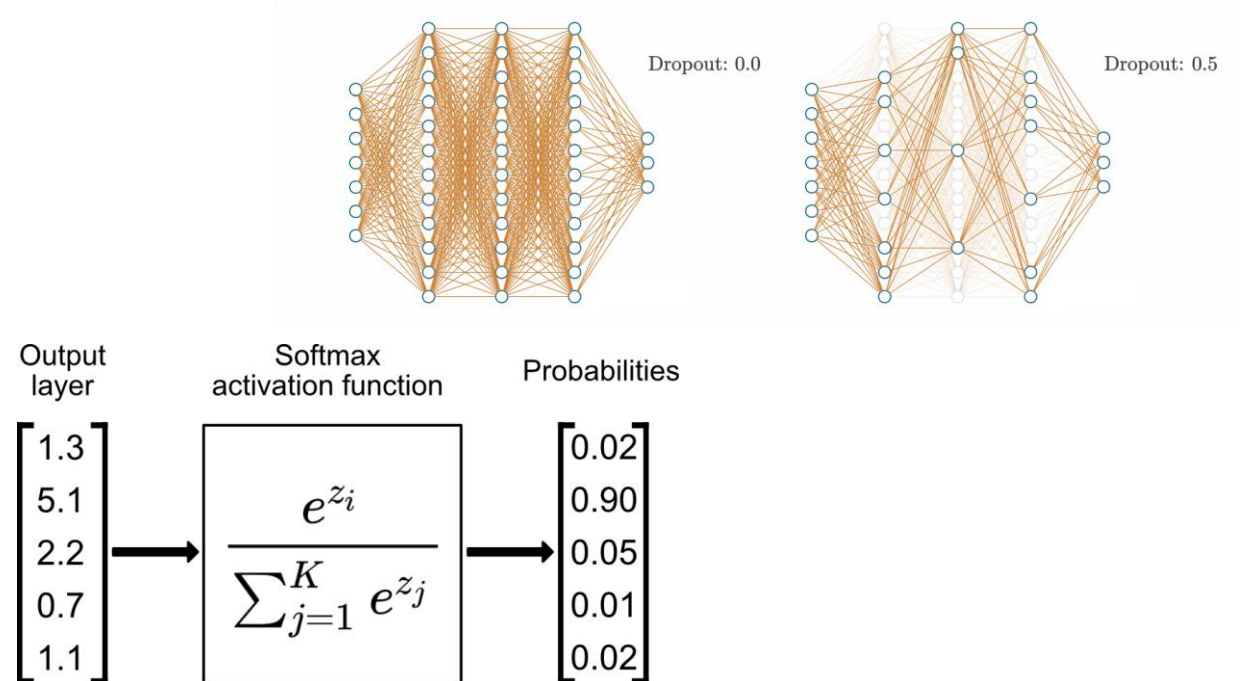
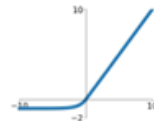


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Computer Vision(CV)

Dataset : mnist, fashion mnist, cifar10, cifar100, ...

Flatten, Dense/Dropout, One-hot encoding, ...

Activation function : relu, softmax, ...

Loss function : mean square error, **categorical_crossentropy**, ...

CV2

NN vs. CNN

$$CELoss = - \sum_{c=1}^N y_c \log P_c$$

y	P	ylogP	-ylogP
0	0.3	0	0
1	0.2	-0.698	0.698
0	0.1	0	0
0	0.4	0	0

y	P	ylogP	-ylogP
0	0.01	0	0
1	0.95	-0.022	0.022
0	0.02	0	0
0	0.02	0	0

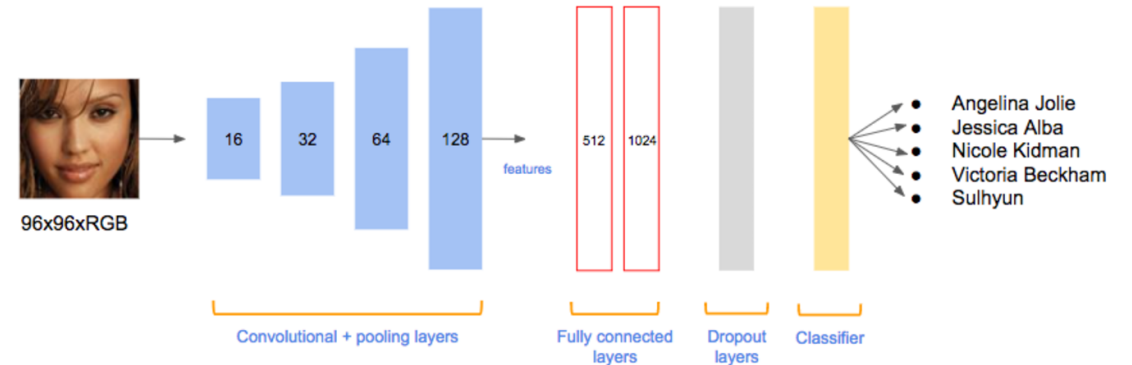
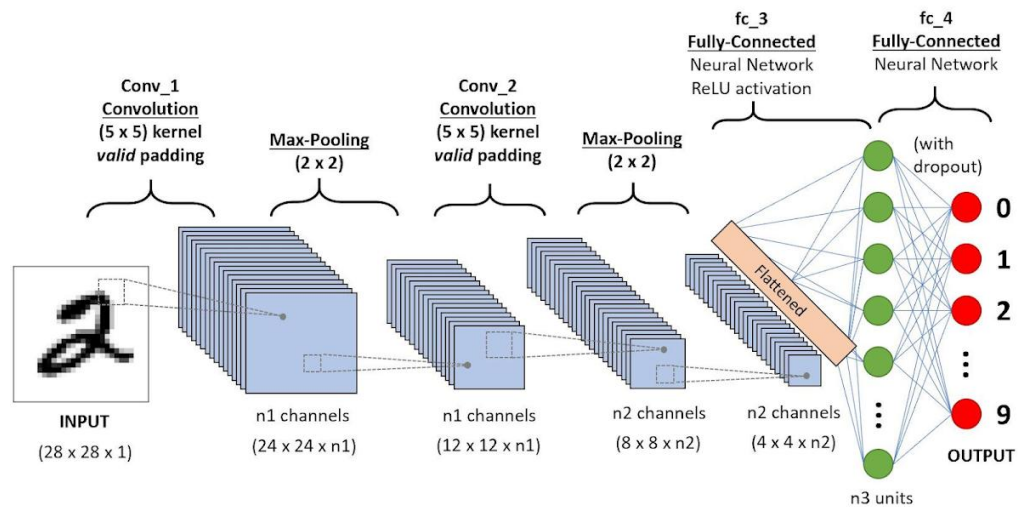
y : one-hot encoded

Computer Vision(CV) : CNN (Convolution Neural Network)

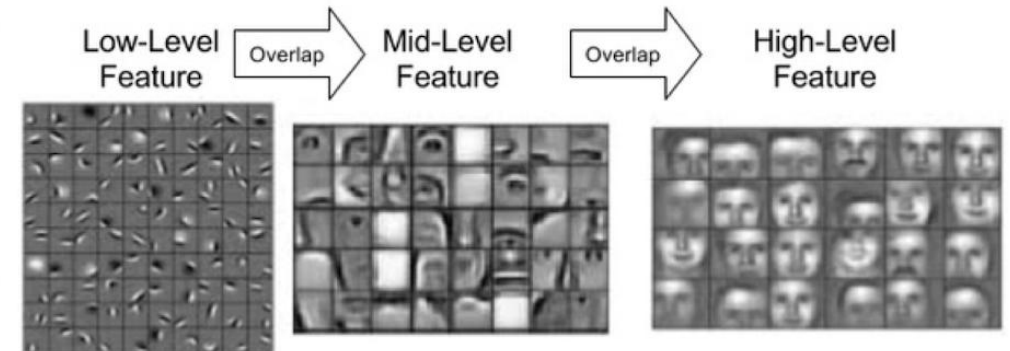
Convolution : padding, stride

Pooling : max, min, average

Filter : Kernel

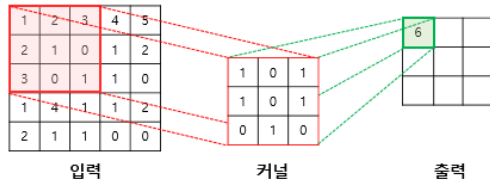


Feature Map in Convolutional Neural Networks (CNN)



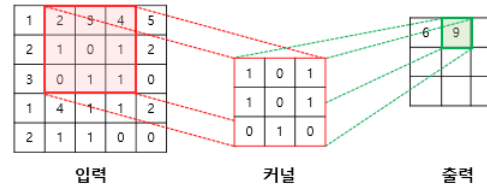
Computer Vision(CV) : CNN (Convolution Neural Network)

1. 첫번째 스텝



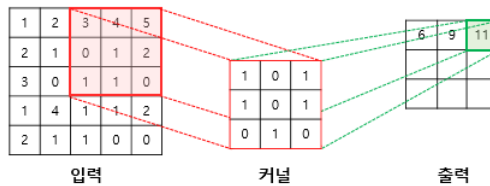
$$(1 \times 1) + (2 \times 0) + (3 \times 1) + (2 \times 1) + (1 \times 0) + (0 \times 1) + (3 \times 0) + (0 \times 1) + (1 \times 0) = 6$$

2. 두번째 스텝



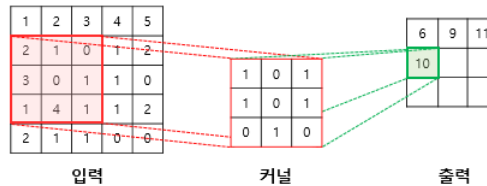
$$(2 \times 1) + (3 \times 0) + (4 \times 1) + (1 \times 1) + (0 \times 0) + (1 \times 1) + (0 \times 0) + (1 \times 1) + (1 \times 0) = 9$$

3. 세번째 스텝



$$(3 \times 1) + (4 \times 0) + (5 \times 1) + (0 \times 1) + (1 \times 0) + (2 \times 1) + (1 \times 0) + (1 \times 1) + (0 \times 0) = 11$$

4. 네번째 스텝



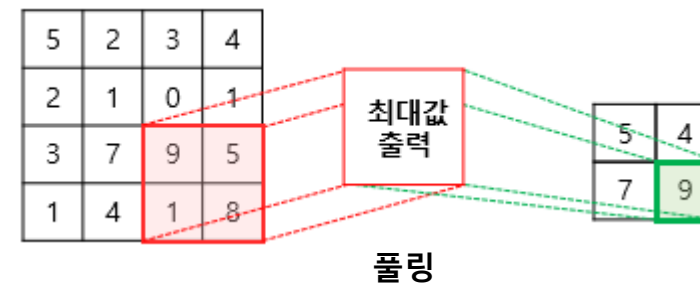
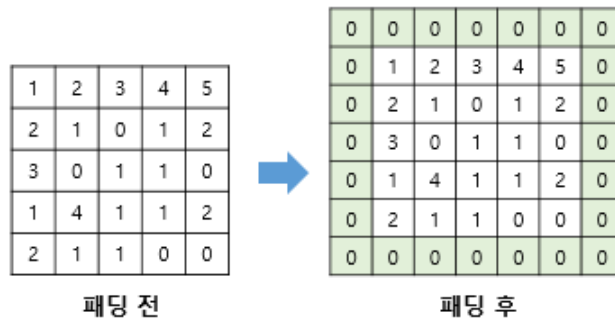
$$(2 \times 1) + (1 \times 0) + (0 \times 1) + (3 \times 1) + (0 \times 0) + (1 \times 1) + (1 \times 0) + (4 \times 1) + (1 \times 0) = 10$$

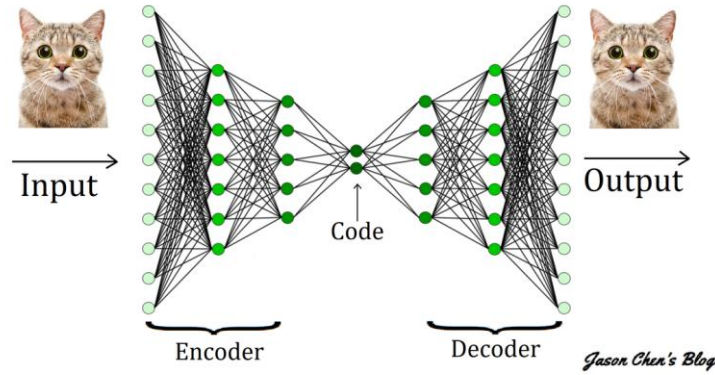
Convolution : padding, stride
Pooling : max, min, average
Filter : Kernel



6	9	11
10	4	4
7	7	4

특성 맵(feature map)





Autoencoder (AE)

Variational Autoencoder (VAE)

GAN (Generative Adversarial Networks)

GPT (Generative Pretrained **Transformer**) : Attention, ...

